Group Completion and the Classification of QCA

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Quantum Spin Systems and Gapped Phases

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Quantum spin system:

- Degrees of freedom on a lattice X (e.g. $\mathbb{Z}^d$)
- Observables form a C^*-algebra $\mathcal{A}(X)$
- A state $\omega : \mathcal{A}(X) \to \mathbb{C}$ encodes expectation values

Gapped state:

- Stable low-energy behavior (energy gap above ground state)

Phase:

- Equivalence class of gapped states under continuous gapped paths

Stacking and Invertible Phases

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Stacking:

$$(\mathcal{A}_1(X), \mathcal{A}_2(X)) \mapsto \mathcal{A}_1(X) \otimes \mathcal{A}_2(X); (\omega_1, \omega_2) \longmapsto \omega_1\omega_2$$

- Induces a commutative monoid structure on phases
- Identity = trivial phase

Invertible phase:

- A phase admitting an inverse under stacking

For each d:

$$\mathcal{I}(\mathbb{Z}^d) = \text{group of invertible gapped phases in dimension } d$$

Kitaev's Conjecture

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Conjecture (Kitaev)

One can construct a space $\mathrm{IP}(\mathbb{Z}^d)$ of invertible states such that

$$\pi_0(\mathrm{IP}(\mathbb{Z}^d)) \cong \mathcal{I}(\mathbb{Z}^d).$$

Moreover, the spaces assemble into an Ω-spectrum:

$$\mathrm{IP}(\mathbb{Z}^d) \simeq \Omega \mathrm{IP}(\mathbb{Z}^{d+1}).$$

Invertible phases $\longleftrightarrow$ Stable homotopy theory

Parametrized Family of States

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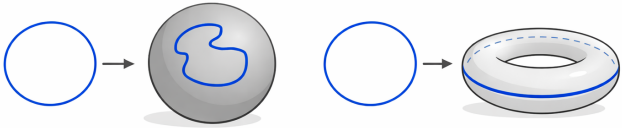

Figure: two families of trivial states

Quantum Cellular Automata (QCA)

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Definition: A QCA in dimension d is a strictly locality-preserving automorphism

$$\alpha : \mathcal{A}(\mathbb{Z}^d) \to \mathcal{A}(\mathbb{Z}^d).$$

Structure[1]**:**

- QCA form a group $\mathcal{Q}^*(\mathbb{Z}^d)$ under composition
- Contains a normal subgroup $\mathcal{C}^*(\mathbb{Z}^d)$ of quantum circuits

$$\mathcal{Q}^*(\mathbb{Z}^d)/\mathcal{C}^*(\mathbb{Z}^d) \quad \text{is abelian (Freedman–Hastings).}$$

[1] After adding arbitary ancilla to stabilize

QCA act on Invertible Phases

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Given an invertible state

$$\omega : \mathcal{A}(\mathbb{Z}^d) \to \mathbb{C},$$

a QCA α produces a new invertible state

$$\omega \circ \alpha.$$

Key fact:

- Circuits $\mathcal{C}^*(\mathbb{Z}^d)$ do not change phase

Hence we obtain a group homomorphism to the image of the trivial phase

$$\mathcal{Q}^*(\mathbb{Z}^d)/\mathcal{C}^*(\mathbb{Z}^d) \longrightarrow \mathcal{I}(\mathbb{Z}^d).$$

QCA probe invertible phases.

The QCA Conjecture

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Conjecture

One can construct a space $\mathrm{QCA}(\mathbb{Z}^d)$ such that

$$\pi_0(\mathrm{QCA}(\mathbb{Z}^d)) \cong \mathcal{Q}^*(\mathbb{Z}^d)/\mathcal{C}^*(\mathbb{Z}^d).$$

Moreover, these assemble into an Ω-spectrum

$$\mathbb{QCA} = \{\mathrm{QCA}(\mathbb{Z}^d)\}_{d\geq 0}.$$

Quantum Cellular Automata $\longleftrightarrow$ Stable homotopy theory

Main Result

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Based on joint work with Mattie Ji, Quantum Cellular Automata: The Group, the Space, and the Spectrum (arXiv:2602.16572).

Theorem (Ji-Y.)

Using algebraic K-theory, we construct intrinsically a space $\mathrm{QCA}(\mathbb{Z}^d)$ such that

$$\pi_0(\mathrm{QCA}(\mathbb{Z}^d)) \cong \mathcal{Q}^*(\mathbb{Z}^d)/\mathcal{C}^*(\mathbb{Z}^d).$$

Moreover, these assemble into an Ω-spectrum

$$\mathbb{QCA} = \{\mathrm{QCA}(\mathbb{Z}^d)\}_{d\geq 0}.$$

Remark

It may be advantageous to view our construction as the Ω-spectrum of *generalized invertible subalgebras*. This perspective is implicit in the proof of Section 4.

Main Result

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In fact, our results go significantly beyond the original setting in two directions.

1. We develop an algebraic theory of QCA over an arbitrary commutative base ring R. For each such ring we construct an associated Ω-spectrum

$$\{\mathbf{Q}(\mathbb{Z}^d)\}_{d\geq 0}.$$

   In fact, we show that $\mathbf{Q}(\mathbb{Z}) = K(\mathrm{Az}(R))$ is the K-theory of Azumaya R-algebras, so the sequence provides a *non-connective delooping* of $K(\mathrm{Az}(R))$.

2. We prove a similar delooping theorem for QCA defined on a class of more general metric spaces X that depends only on the large-scale (coarse) geometry of X. More precisely, we construct an Ω-spectrum

$$\{\mathbf{Q}(X \times \mathbb{Z}^d)\}_{d\geq 0},$$

Another Generalization of the GNVW Index

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Take a QCA over a ring R, define $$\frac{\alpha(\mathcal{A}(\mathbb{Z}\cap(-\infty,\gamma]))}{\mathcal{A}(\mathbb{Z}\cap(-\infty,\gamma])} \in K_0\mathrm{Az}(\mathrm{R}). \tag{1}$$

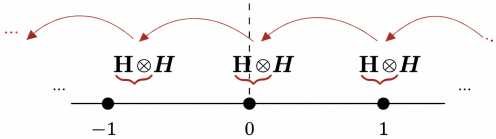

Figure: Real QCA with onsite Hilbert space $\mathbb{R}^4$

This generalizes *the GNVW index* to $K_0(\mathrm{Az}(R))$. For R a field, this is the group completion of the monoid of central simple algebras, which could be worked out using the Wedderburn–Artin theorem.

For example, $R = \mathbb{C}$ this is the group completion of the monoid $\mathbb{Z}_{>0}$ of finite matrix algebras. This recovers the GNVW index in $\mathbb{Q}_>0$.

Other Examples Enabled by the Extension

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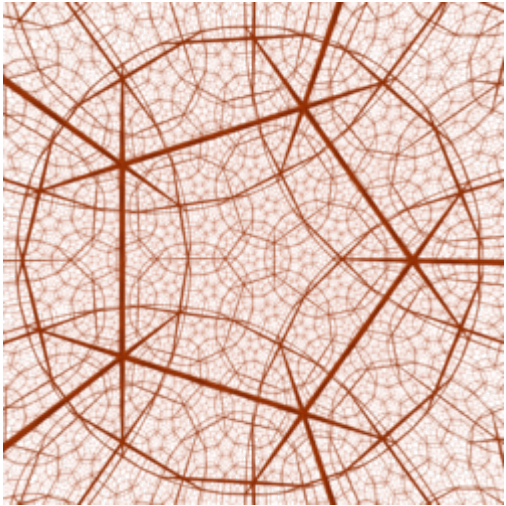

Figure: Hyperbolic lattice

Directly from Groups to Spaces

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First observe the following:

Proposition

$\mathcal{C}^*(\mathbb{Z}^d)$ is a perfect normal subgroup of $\mathcal{Q}^*(\mathbb{Z}^d)$.

Given this, one may glue 2-cells and 3-cells to the classifying space of $\mathcal{Q}^*(\mathbb{Z}^d)$ to obtain a space whose fundamental group is

$$\mathcal{Q}^*(\mathbb{Z}^d)/\mathcal{C}^*(\mathbb{Z}^d).$$

However, I was unable to show that the spaces obtained in this way assemble into a spectrum...

TALKING TO PEOPLE!

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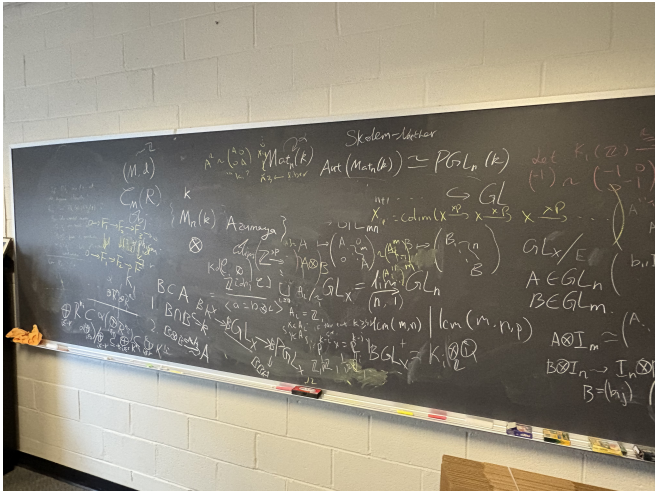

Figure: My Scribbles for Mattie on Kirillov's Blackboard

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From Categories to Spaces

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Strategy:

1. Organize quantum spin systems into a symmetric monoidal category $\mathcal{C}(\mathbb{Z}^d)$:
   - Objects are observable algebras (tensor product of local matrix algebras).
   - Morphisms are locality-preserving isomorphisms
2. Take its group completion to obtain spaces $\mathrm{QCA}(\mathbb{Z}^d)$.
3. Prove the delooping relations:

$$\mathrm{QCA}(\mathbb{Z}^d) \simeq \Omega\mathrm{QCA}(\mathbb{Z}^{d+1}).$$

Group Completion (Technical)

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Spectrum for General QCA

Let $\mathcal{C}$ be a (small) symmetric monoidal category, its classifying space $B\mathcal{C}$ has the natural structure of a homotopy commutative, homotopy associative H-space. In particular, $\pi_0(B\mathcal{C})$ is a commutative monoid.

Definition

The *K-theory space* $K(\mathcal{C})$ is defined as the *group completion* $\iota : B\mathcal{C} \to K(\mathcal{C})$ of $B\mathcal{C}$, which is characterized by the following property

1. $K(\mathcal{C})$ is a group-like H-space. In particular, $\pi_0(K(\mathcal{C}))$ is a group.
2. The induced map $\iota_* : \pi_0(B\mathcal{C}) \to \pi_0(K(\mathcal{C}))$ is the algebraic group completion of the commutative monoid $\pi_0(B\mathcal{C})$.
3. The homology of $H_*(K(\mathcal{C}); R)$ is the localization of $H_*(B\mathcal{C}; R)$ at $\pi_0(B\mathcal{C})$ for any commutative ring R. Here, the homologies are rings because the relevant spaces are H-spaces.

Group Completion (Black Box)

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What this means for us:

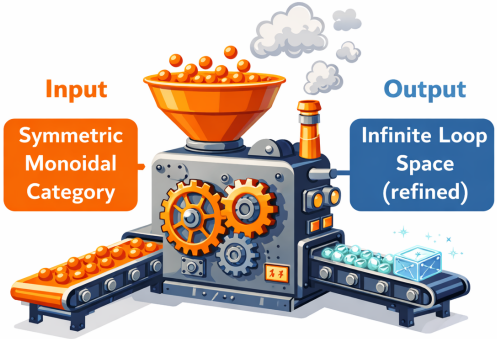

Figure: Think: *monoid* $\longrightarrow$ *group*, but in for categories and spaces.

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The Ω-Spectrum (Section 4)

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Let X be a metric space. We organize quantum spin systems into a symmetric monoidal category $\mathcal{C}(X)$:

- **Objects:** observable algebras (tensor products of local matrix algebras).
- **Morphisms:** locality-preserving isomorphisms.

Apply group completion to get the space $K(\mathcal{C}(X))$. However,

$$K(\mathcal{C}(X)) \not\simeq \Omega K(\mathcal{C}(X \times \mathbb{Z})).$$

Thus, we enlarge $\mathcal{C}(X)$ to $\mathcal{C}'(X)$ by adjoining *(generalized) invertible subalgebras*. Then we prove the key equivalence

$$K(\mathcal{C}'(X)) \simeq \Omega K(\mathcal{C}'(X \times \mathbb{Z})).$$

In particular, the spaces $K(\mathcal{C}'(X \times \mathbb{Z}^n))$ assemble into an Ω-spectrum of invertible subalgebras, which is the Ω-spectrum of QCA shifted by a degree.

The Ω-Spectrum (Section 4)

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We prove the key equivalence

$$K(\mathcal{C}'(X)) \simeq \Omega K(\mathcal{C}'(X \times \mathbb{Z})),$$

where:

- $\pi_0 K(\mathcal{C}'(X))$ is the "Brauer group" of generalized invertible subalgebras,
- $\pi_1 K(\mathcal{C}'(X))$ is the classification group of QCA on X.

For $X = \mathbb{Z}^n$, the spaces

$$K(\mathcal{C}'(\mathbb{Z}^n))$$

assemble into an Ω-spectrum of invertible subalgebras — equivalently, the Ω-spectrum of QCA, shifted by one.

Taking π_0 on both sides yields that over any commutative ring R,

Classification of QCA on $X \times \mathbb{Z}$ $\simeq$ Classification of ISA on X.

Variations on a Theme

Group Completion and the Classification of Quantum Cellular Automata

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Spectrum for General QCA

- **Clifford QCA** The relevant category over any lattice space, together with its group completion, has been studied by Pedersen–Weibel and by Ranicki[2]. A bridge to Clifford quantum cellular automata was outlined in

  "Categorifying Clifford QCA" (to appear in Communications in Mathematical Physics).

- **Fermions QCA.** The same strategy should construct an Ω-spectrum for fermionic (i.e. $\mathbb{Z}/2$-graded) QCA.
- **QCA with symmetries.** Likewise, it should extend to G-equivariant QCA.
- **QCA with tails.** Here we expect to *enrich* the category in topological spaces, i.e. put a metric/norm on the group of QCA so morphism sets become topological spaces.

[2] Haah, CMP (2024)